

Paper Title

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Abstract—

- **Problem:** improve settling prediction for a position-dependent dual gantry without surrendering the meaning of its physical parameters
- **Method:** combine a quasi-LPV first-principles baseline, dynamic parallel LFR augmentation, closed-loop rollout training, and state-level orthogonal projection
- **Evidence:** compare baseline, unconstrained augmentation, orthogonal variants, and a black-box model on the completed validation ladder, including relevant motion and settling where available
- **Main result:** report predictive improvement together with the measured limits from short-horizon training, inactive added states, and non-orthogonal true missing dynamics
- **Conclusion:** state when the proposed decomposition is useful and which conditions prevent interpreting it as true parameter recovery

[TODO: Write the numerical abstract only after the final simulation and Telica result tables are frozen.]

Index Terms—grey-box identification, LPV systems, model augmentation, orthogonal projection, dual gantry

I. INTRODUCTION

A. Industrial motivation and modelling gap

- Settling prediction limits throughput in coupled high-precision gantries, so model error has a direct operational cost [1]
- The concrete digital-twin use case is to evaluate relevant ASMPT motion and a changed controller before deployment on the machine
- The physics model exposes masses, stiffnesses, damping, and coupling, but omits position-dependent flexible dynamics that matter for prediction
- A black-box model offers flexibility but weakens extrapolation arguments and removes parameters useful for control design, digital twins, and health monitoring
- Grey-box dynamic augmentation is the target compromise: preserve the baseline and learn only the missing state dynamics [2]

[TODO: Which industrial benefit should lead: settling prediction, controller design, offline commissioning, or condition monitoring?]

ASMPT develops semiconductor assembly and packaging equipment, including high-precision motion systems for advanced packaging. Advanced packaging refers to combining multiple semiconductor functions within a single package,

for example through heterogeneous integration of different die types such as logic and memory, and through hybrid bonding, in which dies are directly joined at very fine pitch. [TODO: Shorten to a clause inside the previous sentence (page budget).] Such technologies place strict demands on alignment accuracy, yield, throughput, and manufacturing cost [3], [4]. [TODO: Only partly supported. IRDS covers heterogeneous integration, cost and yield, but never mentions hybrid bonding and its throughput means memory bandwidth. Workman covers sub-micron alignment and yield. Neither supports machine throughput: drop it here or add a source. Workman bib entry: authors incomplete, pages 2071 to 2077 missing.]

This project focuses on a dual-gantry motion system [TODO: Tense and terminology: project becomes paper.] in which two parallel linear actuators support a common beam carrying the payload. The shared beam couples the actuator dynamics, so motion of one actuator influences the other, and the system exhibits position-dependent behaviour and flexible dynamics [1], [5]. [TODO: Teo (abstract only) supports coupling and payload-position dependence, not flexibility. Flexibility rests on Garcia Sec. 2 (flexible joints kb1, kb2): consider flexible joint dynamics.] In such systems, settling behaviour is a key performance metric. After each motion, the system must wait for residual oscillations to decay before the next placement can begin [6], so settling time directly affects throughput. Furthermore, cross-coupling means that the motion of one gantry can disturb the other, making accurate prediction relevant not only for settling but also for tracking behaviour during simultaneous motion.

Improving settling performance therefore requires an accurate representation of the underlying plant dynamics. ASMPT currently captures part of the position dependence of the dual-gantry system using a baseline model based on frequency-response measurements at multiple operating points with interpolation across the operating space. [TODO: No written source: confirm with Quinten or Jasper. The paper does not compare against this model, so do not imply it.] However, this baseline captures only part of the position dependence and does not systematically account for cross-coupling between the gantries. [TODO: Same confirmation as above.] A more accurate plant model is therefore needed. Such models are not only valuable for settling prediction, but also for digital-twin development, offline evaluation, control design, and condition monitoring. [TODO: Lead with the controller-evaluation digital twin: test a changed controller on the model before the machine

Identify applicable funding agency here. If none, delete this.

(Quinten, 18 Sep). Condition monitoring is not evaluated; its supporting sentence was removed.] Digital twins can support offline simulation of semiconductor manufacturing equipment and help optimize operating parameters before testing changes on the real machine, reducing costly trial-and-error [7]. [TODO: Only partly supported (abstract only). Hwang optimises operating parameters of wafer-processing cluster tools by digital-twin simulation; before testing on the real machine and trial-and-error are not in the source.]

White-box models are attractive because they reflect the physics of the system, which makes them interpretable. First-principles (FP) models are a specific type of white-box model, derived from physical laws and described by quantities such as mass, stiffness, and damping. However, purely physics-based models do not always capture all behaviour needed for accurate prediction across the operating range [8]. For the dual-gantry system, this limitation is also evident in the first-principles model structure of García-Herreros et al. [1], which is only able to capture part of the position-dependent behaviour and cross-coupling. [TODO: Not what Garcia says. Garcia Sec. 2 deliberately neglects cross-arm vibration on the flexible plates, detent forces, and friction variation along the stroke. Our LPV-LFR baseline captures the M(Y) position dependence exactly; the missing dynamics here are the hidden absorber mode and, in the friction twin, Coulomb friction in the truth only (D-204).]

Purely data-driven models such as neural networks or Gaussian processes can describe measured behaviour well, but they give less physical insight, are often harder to trust [TODO: Grammar: an and is missing after insight.] at operating conditions not covered by the training data [8]. Typically these models also require large amounts of data and significant computational power to train. [TODO: Uncited. Large data requirements: Willard Sec. 1. Computational power: no source found, drop or cite. Verified alternative: black-box models spend effort rediscovering dynamics already known from first principles (Hoekstra 2025 EJC p. 1).] For the plant model of the dual-gantry system, both predictive accuracy and physical interpretability are required. However, neither a purely physics-based nor a purely data-driven model is sufficient on its own [8], [9]. [TODO: Cite Willard only: Noel never argues that pure physics models are insufficient. Willard bib: ACM Computing Surveys 2022 (issue 2023), not 2020; month field wrong.] Grey-box modeling is the logical choice, because it is able to combine a physically meaningful baseline model needed for interpretation with data-driven flexibility needed for accurate prediction [9]. [TODO: Noel supports only grey-box combining physical structure with data-driven flexibility. Name model augmentation explicitly and cite Hoekstra 2025 EJC and Hoekstra 2026.] However, related work at ASMPPT shows that preserving physical interpretability during model updating remains an open challenge [10]. [TODO: Supported by Kessels thesis Ch. 6 recommendations (PDF p. 222): augmentation can compensate for incorrect FP parameters, compromising their interpretability, and orthogonality methods exist only for linear-in-the-parameter cases. The thesis says industrial wire bonder and never names ASMPPT: confirm or reword.]

Learning-based hybrid modeling has already been demonstrated on real industrial systems [11]. [TODO: Supported (Retzler 2024, industrial robot, measured data).] However, a key gap remains: there is no validated grey-box modeling framework for the dual-gantry system that simultaneously addresses coupled multi-input multi-output (MIMO) dynamics, position-dependent behaviour, and preserves physical interpretability. [TODO: Rewrite the gap as a verified intersection. Hoekstra 2025/2026: dynamic LFR augmentation, no LPV scheduling, no closed-loop training. Kessels: closed-loop-trained augmentation on an industrial motion system, but no LFR or LPV, frozen FP parameters, no orthogonality, so closed-loop training alone is not the novelty. Gyrook OBC 2026: discrete-time input-output models only; its conclusion names state-space without full-state measurement as future work. Validated must be scoped to simulation.] This project therefore aims to develop and validate such a framework [TODO: Tense and scope: this paper develops and evaluates in simulation.] so that the plant model becomes more accurate for settling prediction while remaining physically meaningful.

B. Research question and scope

- Main question: how should a dual-gantry physics model be augmented to improve settling prediction while retaining physical parameter interpretability?
- Aspect 1 asks whether the payload-position scheduling should remain explicit in a quasi-LPV baseline rather than frozen LTI models
- Aspect 2 asks whether a dynamic parallel augmentation with additional states captures flexible cross-coupling absent from the baseline
- Aspect 3 asks what state-level orthogonality can and cannot guarantee during joint estimation
- Aspect 4 asks whether the model generalises across operating points and motion profiles relative to a black-box state-space model
- Scope separates interpolation, transfer to ASMPPT-relevant motion, controller transfer, operating-range extrapolation, and any unavailable hardware evidence

[TODO: Does the approved research-plan wording need to remain verbatim, or may the final question reflect the closed-loop and OBC developments?]

The main research question is formulated as follows:

For the dual-gantry system, how should physics-based plant models be augmented to improve parameter interpretability and settling prediction accuracy, particularly of cross-coupling and position-dependent dynamics? [TODO: Decision A open: approved wording says improve parameter interpretability, which augmentation cannot do; candidate minimal edit is augmented from closed-loop data to improve settling prediction accuracy while preserving parameter interpretability.]

To answer this research question, the following four aspects are investigated:

1) Position-dependent baseline modeling

In what form should the baseline model be represented for subsequent augmentation, and does position-dependent (LPV) formulation improve prediction compared to position-independent (LTI) alternatives? [TODO: LPV versus frozen LTI exists only as a baseline check (lpv_lfr_baseline/scripts/validate_lfr.py), not in the final evaluation; Quinten wants LTI as second baseline.]

2) Augmentation structure

What augmentation structure can effectively capture dynamics not modeled in the physics-based baseline model, in particular flexible cross-coupling dynamics?

3) Interpretability preservation

How can parameter interpretability be maintained during data-driven augmentation, compared to unconstrained approaches? Here, interpretability means that the physical parameters of the baseline retain values close to their physically meaningful initialization after augmentation. [TODO: Definition no longer matches the method. The OBC arms start from 10 percent detuned combinations with the prior disabled (D-193) and judge recovery of the ten identifiable combinations. Staying near initialization is what a Lambda penalty trivially gives.]

4) Model generalization

How well does the augmented model generalize to (i) held-out operating points representative of actual industrial operating conditions and (ii) unseen motion profiles relevant to the industrial use context, and how does its prediction accuracy compare to a black-box state-space model trained on the same data? [TODO: Not yet true. The D-196 black-box arm runs on the MA_FRAC 0.10 dataset, not the production ma50 data. Unseen motion profiles partly covered by the Telica cycloidal records (D-206). Controller transfer is not implemented. Align with the validation ladder in Writing/README.md.]

C. Contributions and paper organisation

- Derive a compact continuous-time LPV-LFR realisation of the dual-gantry baseline with rational dependence on payload position
- Extend dynamic parallel LFR augmentation to a closed-loop, self-scheduled state-space setting with learned additional states
- Construct state-level orthogonality for the identifiable physical-parameter tangent space and quantify its true-parameter recovery condition
- Evaluate predictive fit, settling, controller transfer, added-state contribution, interpretability, and black-box performance under matched data
- Paper organisation: baseline, augmentation, OBC, setup, results, discussion, conclusion, then supporting derivations
- Figure plan: no dedicated introduction figure; use the limited figure budget where the baseline, training interconnection, and OBC need visual explanation

[TODO: Confirm whether references and appendices are excluded from the 12-page IEEE limit and whether TU/e

requires front matter beyond the IEEE paper.]

II. DUAL-DRIVE GANTRY SYSTEM AND PHYSICS BASELINE

A. System and first-principles baseline

The physics baseline used in this work is based on the lumped-parameter model of the dual-drive gantry developed by García-Herreros et al. [1]. The gantry has two actuators that carry a cross-arm and a third actuator that moves the payload along the arm (Fig. 1). [TODO: Does this make sense to put here?:] Dynamics not represented by the baseline are treated as model discrepancy and addressed by the augmentation in Section III. Define the actuator forces $u_{\text{act}} = [F_{X1} \ F_{X2} \ F_Y]^T$, the measured positions $q_{\text{act}} = [X_1 \ X_2 \ Y]^T$, and the generalized coordinates $q = [X \ \Theta \ Y]^T$. Let L_b denote the cross-arm length between the two flexible joints. The coordinate and force transformations are

$$q = P^{-T} q_{\text{act}}, \quad u = P u_{\text{act}}, \quad (1)$$

$$P = \begin{bmatrix} 1 & 1 & 0 \\ L_b/2 & -L_b/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where u denotes the generalized force. The generalized coordinates are defined as $X = (X_1 + X_2)/2$ and $\Theta = (X_1 - X_2)/L_b$, while Y is unchanged. These coordinates avoid a closed kinematic-chain description and make the coupling caused by the nonuniform load distribution explicit [1].

Following García-Herreros et al. [1], the baseline uses $\cos \Theta \approx 1$ and $\sin \Theta \approx 0$ and assumes that the Coriolis and centripetal force term is negligible compared with the inertia, damping, and stiffness forces. Under these assumptions, the baseline equations of motion are

$$M(Y)\ddot{q}(t) + C\dot{q}(t) + Kq(t) = P u_{\text{act}}(t), \quad (2)$$

with

$$M(Y) = \begin{bmatrix} m_1 + m_2 + m_b + m_h & \frac{L_b}{2}(m_1 - m_2) - m_h Y & 0 \\ \frac{L_b}{2}(m_1 - m_2) - m_h Y & J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2) + m_h(d^2 + Y^2) & -m_h d \\ 0 & -m_h d & m_h \end{bmatrix}, \quad (3)$$

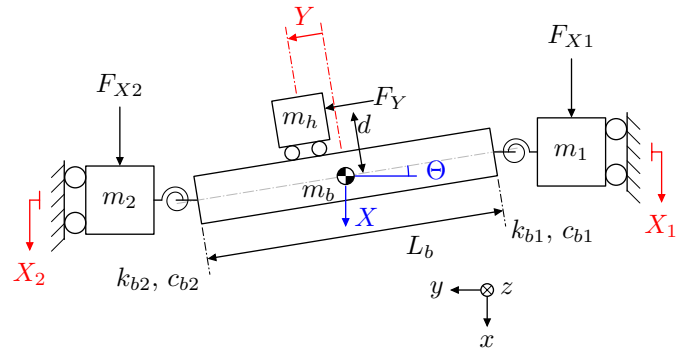


Fig. 1: Lumped-parameter gantry schematic adapted from the mechanical arrangement in [1]. Red labels denote actuator coordinates and blue labels generalized coordinates.

$$C = \begin{bmatrix} c_{g1} + c_{g2} & \frac{L_b}{2}(c_{g1} - c_{g2}) & 0 \\ \frac{L_b}{2}(c_{g1} - c_{g2}) & c_{b1} + c_{b2} + \frac{L_b^2}{4}(c_{g1} + c_{g2}) & 0 \\ 0 & 0 & c_y \end{bmatrix}, \quad (4)$$

$$K = \begin{bmatrix} 0 & 0 & 0 \\ 0 & k_{b1} + k_{b2} & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The masses, flexible-joint stiffness and damping coefficients, and geometric dimensions are shown in Fig. 1. Here, J_b and J_h denote the yaw inertias, while c_{g1} , c_{g2} , and c_y denote the actuator damping coefficients. The nominal values are taken from [1, Table I] and listed in Appendix A-B [TODO: Not sure if to add this, depends on real data or not; they were not identified on the present Telica machine.] To prevent the definition of q from being parameter-dependent, the cross-arm length L_b is fixed at the nominal value listed in Appendix A-B rather than treated as a free model parameter. [TODO: Determine and justify the operating range of Y ; distinguish the mechanical stroke, simulated-data range, and range covered by the measured ILC records.]

B. Self-scheduled LPV-LFR representation

The position dependence of (3) can be written as

$$M(Y) = M_0 + Y M_1 + Y^2 M_2, \quad (5)$$

where M_0 , M_1 , and M_2 are constant. The measured payload position Y is used as the scheduling variable. Because Y is also a model state, the representation is self-scheduled, or quasi-LPV [12].

Define the net generalized force as

$$f_{\text{net}} = P u_{\text{act}} - C \dot{q} - K q. \quad (6)$$

Solving (2) for $\ddot{q}$ introduces the rational dependence $M(Y)^{-1}$. Rather than forming this inverse explicitly, the LFR is constructed from the mass equation by introducing

$$w_1 = Y \ddot{q}, \quad w_2 = Y w_1 = Y^2 \ddot{q}. \quad (7)$$

The resulting Y -independent relation is

$$M_0 \ddot{q} + M_1 w_1 + M_2 w_2 = f_{\text{net}}. \quad (8)$$

Both products in (7) apply the same scalar Y to a three-dimensional signal. Defining

$$z = \begin{bmatrix} \ddot{q} \\ w_1 \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \quad (9)$$

their stacked relation is

$$w = \underbrace{\begin{bmatrix} Y I_3 & 0 \\ 0 & Y I_3 \end{bmatrix}}_{\Delta(Y) = Y I_6} z. \quad (10)$$

Thus, all dependence on Y is contained in the scheduling block $\Delta(Y)$ [13]. With $x = [q^\top \dot{q}^\top]^\top$, the remaining relations define the constant block

$$\begin{bmatrix} \dot{x} \\ z \\ q_{\text{act}} \end{bmatrix} = \underbrace{\begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & 0 & 0 \end{bmatrix}}_G \begin{bmatrix} x \\ w \\ u_{\text{act}} \end{bmatrix}, \quad (11)$$

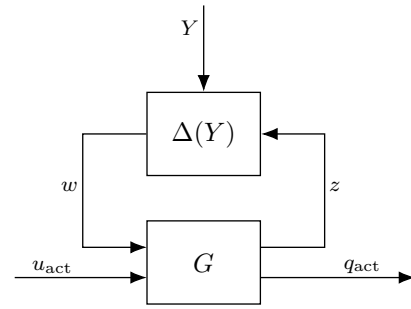


Fig. 2: Self-scheduled baseline interconnection: the LTI part G is closed by $\Delta(Y) = Y I_6$, where Y is a component of the state.

The state is expressed in generalized coordinates, whereas the input u_{act} and output q_{act} remain in actuator coordinates to match the applied forces and measured positions. The transformations in (1) are included in G , whose matrices are listed in Appendix A-B.

Closing the loop in (8) gives

$$M_0 \ddot{q} + Y M_1 \ddot{q} + Y^2 M_2 \ddot{q} = f_{\text{net}}, \quad (12)$$

which, by (5) and (6), is equivalent to (2).

The internal loop is well posed if and only if $M(Y)$ is nonsingular. For positive component masses and yaw inertias, $M(Y)$ is positive definite for every $Y \in \mathbb{R}$; hence, the nominal interconnection is globally well posed. The proof is given in Appendix A-B. The continuous-time model is integrated using the fixed-step RK4 scheme specified in Section V.

C. Physical parameters and identifiable combinations

- Distinguish the fourteen raw parameters from the ten combinations identifiable through the gantry equations
- Report recovery in the reduced combinations and state explicitly whether each experiment trains raw or reduced coordinates
- Show the Jacobian from raw parameters to identifiable combinations; naive parameter merging is not a valid recovery argument
- Use exact-model parameter recovery only as an internal implementation/initialization check; do not allocate a main result to it
- Use relative combination error as the interpretability metric and disclose the special normalisation used for the small signed mass-difference combination
- State that L_b is fixed in the transformation P and is therefore not identified.

[TODO: Should the raw-to-reduced Jacobian be printed in the appendix or shown only as rank and singular values?]

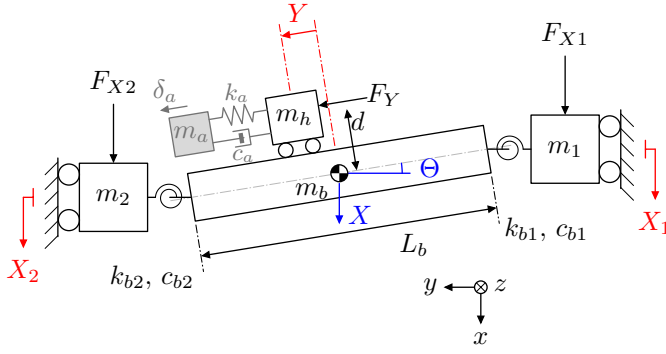


Fig. 3: Simulated ground-truth system: the baseline of Fig. 1 with a payload-mounted absorber. The light-gray elements (m_a , k_a , c_a , δ_a) are absent from the baseline.

The fourteen raw physical parameters enter the model only through

$$\begin{aligned} \varphi &= [k_{b,\Sigma}, c_{g1}, c_{g2}, c_y, c_{b,\Sigma}, m_h, m_\Sigma, m_\Delta, J_{\text{eff}}, d]^\top, \\ k_{b,\Sigma} &= k_{b1} + k_{b2}, \quad c_{b,\Sigma} = c_{b1} + c_{b2}, \\ m_\Sigma &= m_1 + m_2 + m_b, \quad m_\Delta = m_1 - m_2, \\ J_{\text{eff}} &= J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2). \end{aligned} \quad (13)$$

The raw-to-model Jacobian has rank ten, so four raw parameter directions do not affect the dynamics. Parameter recovery is therefore assessed using φ ; individual raw splits are not interpreted without an explicit gauge choice. The mapping, null directions, and admissibility condition are derived in Appendix A-B. Structural identifiability does not guarantee that a finite dataset excites all ten directions; practical recoverability is assessed in Section V. **[TODO: Admissibility is checked at construction and evaluation but is not constrained during training. Verify that every reported checkpoint passes and decide whether a training-time guard is required.]**

III. DYNAMIC LPV-LFR AUGMENTATION

A. Parallel augmentation with additional states

- Use a dynamic parallel LFR so the learned component adds missing dynamics while the physics state update remains explicit [2]
- Partition the model state into six physical states and learned additional states, with the ANN writing both partitions and the output reading physical states only
- Explain why dynamic states are needed for an omitted resonance and why static output correction or parameter re-estimation cannot create new poles
- Define the state router, ANN inputs, zero output readout at initialisation, and the recurrent path by which additional states affect later physical outputs
- The additional-state Jacobian blocks replace any claim that the learned states possess separate explicit A and B matrices

- Figure (existing, major revision): simplify jan-blockscheme-v5 to baseline, ANN, router, physical states, additional states, and output; remove experiment-specific blocks unless they remain in the final model
- Figure change: separate the continuous-time derivative model from one explicit RK4 block at T_s , and label the state register by $\hat{x}_{k+1} \rightarrow \hat{x}_k$ so it is not mistaken for a plant delay

[TODO: How many added states can be justified from the omitted-mode physics rather than selected only by validation performance?]

B. Encoder and state initialisation

- Initialise each truncated rollout with an encoder using past input-output samples and a model-based linear map for the physical rows [14]
- Why not use the positions and finite difference directly for the known initial states? Clearly state why we have chosen for this.
- Why are we augmenting the positions while directly measured? state the reason for this choice.
- Acknowledge that the published linear initialisation does not determine the additional-state rows, which begin from an unsupported random map
- Explain the zero-gradient path into added states at initialisation and the two zero gates that delay their use
- Use the exact simulated state for encoder-only diagnosis, but never as information available to the deployable model
- Compare encoder-history lengths and ANN capacity because the chosen windows determine reconstructability and startup transients

[TODO: Should the trained encoder remain part of the final model, or should a longer measured-history observer be treated as a separate deliverable?]

C. Closed-loop rollout objective

- Motivate the formulation from the industrial constraint that relevant machine data are obtained under feedback; the integrating axes explain the resulting long-horizon sensitivity rather than the existence of the problem
- Drive the model with $\hat{u}_k = u_k + C_{fb}(y_k - \hat{y}_k)$ so the simulated linear controller reacts to model error as the machine controller did
- Derive the residual form from controller linearity only and cite direct closed-loop augmentation as the published fallback for nonlinear controllers [10]
- Closed-loop training avoids reconstructing plant and controller states before every window because their prior response is already carried by recorded input
- Mark the departure from the research plan's open-loop free-run training and explain that feedback can hide model error and orthogonality overlap
- Train on short normalised windows and score the complete window without burn-in, but select and report on long free runs to expose integrated velocity bias

- Define controller-transfer validation separately: identify with the training controller, then evaluate a frozen credible alternative controller without calling that change plant extrapolation by itself
- Figure (existing, revise): retain the three-part derivation in `closed-loop-form-v4`, simplify the explanatory text, and make the final residual rollout equation the visual conclusion
- Figure (existing template, rebuild): replace the open-loop claim in `training-objective-v1` with a time strip for encoder history, the fully scored closed-loop rollout, and long-horizon validation

[TODO: Can ASMPT confirm saturation, anti-windup, and within-record controller scheduling, the three assumptions that determine whether the residual form is valid?]

D. Why augmentation rather than competing residual models

[TODO: Not sure if I should state that here, whether this section is appropriate.]

- Parameter refitting cannot represent a missing state and a static correction cannot reproduce flexible memory
- A residual FRF or fitted mass line is a useful linear benchmark but does not supply one nonlinear position-dependent state-space model for arbitrary trajectories
- Basis-function augmentation would improve transparency but requires choosing the missing physics class in advance and scales poorly across coupled effects
- Kessels supplies the closest closed-loop state-extension precedent, while this work adds LPV-LFR structure and an interpretability safeguard [10]
- The black-box comparator tests whether retaining the physics baseline adds value beyond model capacity alone

[TODO: Which competing method deserves an implemented benchmark beyond the required black-box model: residual BLA, basis functions, or parameter-only refitting?]

IV. INTERPRETABILITY BY ORTHOGONAL CONSTRUCTION

A. Negation and the protected tangent space

- Joint estimation is non-unique when the ANN can reproduce or cancel changes reachable by physical parameters [?], [15]
- Define interpretability as recovery of the ten identifiable physical combinations, not visual plausibility of individual raw parameters
- Stack one-step sensitivities $J = \partial f_\theta / \partial \theta$ over a data-derived reference set in normalised state coordinates
- Protect the rank-ten tangent space and compare the optional affine-offset column as a separate arm rather than silently changing the definition
- Report tangent-space rank, condition number, and correlated sensitivity columns before interpreting parameter recovery
- Figure (new): geometric decomposition $g = P_J g + (I - P_J)g$ showing the baseline-aligned correction removed and the orthogonal correction retained

- Use the same figure to show that local tangent separation does not imply global, trajectory-level, or latent-state uniqueness

[TODO: Should the thesis call the affine arm a stricter non-negation definition or only a sensitivity analysis around the chosen tangent definition?]

B. State-level orthogonal-by-construction map

- Project the ANN write into the physical rows with $I - JJ^+$, eliminating the penalty weight used by soft orthogonality regularisation [15]
- Freeze the reference basis within each epoch and refresh it as the physical parameters move, keeping the recurrent transition causal
- Differentiate through the projection coefficient while treating the stacked basis as constant during each optimisation interval
- Orthogonality applies to the physical one-step write; additional-state rows are orthogonal to the baseline by the baseline's zero extension
- Additional states still propagate into later physical outputs, so one-step state orthogonality does not imply trajectory-output orthogonality
- Separate the no-projection, ten-column tangent, and eleven-column affine arms under matched data, initialisation, and optimisation budgets
- Figure change: add a small state-partition inset showing projection on physical rows, unrestricted additional-state rows, and their delayed influence through recurrent propagation rather than a direct output path

[TODO: Is one-step state orthogonality the intended thesis contribution, or must the trajectory-level penalty also appear despite its unresolved weight selection?]

C. Condition for true parameter recovery

- OBC makes the local decomposition unique, but true recovery additionally requires the real missing dynamics Δ^* to satisfy $J^\top \Delta^* = 0$
- Derive the least-squares parameter bias $\hat{\theta} - \theta^* = J^+ \Delta^*$ and define $\rho^* = \|JJ^+ \Delta^*\| / \|\Delta^*\|$
- Measure $\rho^* = 0.205$ for the hidden absorber, so orthogonality fails on the current excitation and zero parameter error is not the valid target
- The predicted combination-error floor is 4.96%, dominated by a mass-imbalance direction sharing the absorber's equilibrium offset
- Exact pointwise orthogonality is impossible for a nonzero force on this system; only excitation-averaged orthogonality can be designed

[TODO: Should the proposed orthogonal multisine design be a future-work result, or must it be run before claiming a constructive remedy?]

V. EXPERIMENTAL SETUP

A. Simulation benchmark and excitation

- Generate closed-loop records from the physics baseline plus a truth-only flexible absorber, retaining exact states solely for diagnostics
- Use multisine excitation that covers the absorber pole and anti-resonance while respecting position, velocity, yaw, and force limits
- Assign independent random multisine phases to training, validation, and test records so equal spectra do not duplicate trajectories
- Include multisine-free cycloidal motion profiles matching the measured machine motion to test the operational distribution shift
- Use only the corrected all-channel Coulomb dataset matched to the frictionless excitation; the earlier mismatched-amplitude merge is superseded
- The set-valued friction law requires a velocity threshold below which a rail is treated as stuck. This is set to a physical breakaway velocity, $v_{\text{brk}} = 2.25 \cdot 10^{-3}$ m/s, taken from Lee et al. [16], Table 5, identified on a THK SSR20XW linear-motion guide. Reported Stribeck velocities for lubricated guideways span 10^{-5} to 10^{-2} m/s [16, Sec. 5.2], so the chosen value sits mid-range **[TODO: Conflicts with D-204, where the stick band is $V_{\text{EPS}} = (cc1+cc2)/m_{\text{total}} \cdot ts$ labelled THEORY: leine1998, about $3.3 \cdot 10^{-5}$ m/s before the 3 to 10x margin. Confirm which threshold the reported runs use, and whether the second citation (Stribeck range) is the same paper.]**
- State that output noise is currently disabled; add it only if a level and measurement location can be estimated from Jasper, ILC data, or stationary records
- State the sample rates, record lengths, number of windows, and total independent motion time rather than quoting overlapping-window count as data volume
- Decision (RK4 value, moved here from Section II): one reproducibility sentence giving the RK4 step, equal to the sampling period after downsampling the 20 kHz records to 4 kHz, with one substep (entry script: `fs_new=4000, up_sample=1`). Confirm against the `config.json` of each reported run, since earlier checkpoints may have used another `up_sample`
- Figure (reuse, no separate asset): let the rebuilt training-objective time strip also identify independent phases, short training windows, encoder history, and the long evaluation horizon

[TODO: Is the hidden linear absorber sufficient simulation complexity, or is a truth-only cubic spring required to demonstrate nonlinear augmentation?]

B. Real Telica data and split

- Confirm with Jasper and Dragan which reconstructed ETEL setpoint-generator or ILC-reference motions are op-

erationally representative, including minimum/maximum move, long stroke, acceleration, and dwell

- Use the supervisor-defined split of 11 training, two validation, and two test operating points at 20 kHz
- Convert rail positions and currents to logical metres and Newtons; form velocity states by central differences because direct velocity is not logged
- Keep all iterations at one operating point in the same split to prevent identical reference trajectories leaking across sets
- Describe the X and Y operating-point grid and show whether training covers the interpolation region occupied by both test points
- Use iteration 0 for controller checks and later ILC iterations for distinct feedforward conditions without treating them as independent trajectories
- Mark real-data training and held-out evaluation as planned until a final reproducible run and force-calibration convention exist
- Figure (new): 2D operating-point grid with train, validation, and test locations, operating-range boundary, and trajectory classes; add a 3D or pairwise state-input coverage view only if it reveals coverage hidden by the 2D map

[TODO: Freeze the approved operational trajectory set with Jasper and Dragan; should direct drive velocity be requested, and does a third coverage coordinate add information beyond the 2D grid?]

C. Training protocol, comparators, and metrics

- Match records, neural seeds, windows, optimiser budget, and checkpoint criterion across baseline-only, unconstrained, tangent-OBC, affine-OBC, and black-box arms; treat multisine phase seeds separately
- Use physical RMS as the primary candidate metric for tracking and a fixed 200–400 ms post-move settling interval; use NRMS, BFR, or spectra only for a distinct supporting question
- Report physical combination error, projection overlap, added-state ablations, and encoder-init sensitivity alongside predictive metrics
- Decision (moved from Section II-C): joint-estimation arms start from the nominal φ of the reference implementation detuned by 10% per combination (`combo_init_detune`, HEURISTIC, D-191), with the nominal-parameter prior disabled (D-193); state it as a run setting with its reason
- Practical identifiability (referenced from Section II-C): report the rank of the stacked one-step transition Jacobian with respect to φ on the training data (rank 10, nullity 4 in θ per the reduced-block docstring), from a saved artefact rather than the docstring
- Give the black-box model validation-selected capacity and horizon so it is a capability comparison rather than an artificially weakened baseline

- Evaluate held-out position/phase interpolation, ASMP-relevant multisine-free motion, controller transfer, and operating-range extrapolation as separate rungs
- Predeclare expectation, acceptance criterion, verdict, and mechanism for every result rather than presenting isolated scores

[**TODO:** Decide the primary metric, credible controller-transfer arm, and feasible neural-seed count; do not pool initialization, noise, phase, and operating-point uncertainty.]

D. Reproducibility and implementation checks

- Verify controller reconstruction, coordinate transforms, normalisation, RK4 transition, and LFR equivalence before training
- Use the same controller, data conversion, and rollout entry point for training, validation, and checkpoint selection
- Record configuration, software precision, seed, dataset identifier, and checkpoint provenance for every reported arm
- Keep oracle states and truth parameters out of the trainable pipeline and label every diagnostic that uses them
- Figure (existing, appendix candidate): use `controller-chain-v1` only as an audit trail for controller reconstruction and discretisation, not as a main method figure

[**TODO:** Which implementation checks belong in the main paper, and which should move to the appendix or repository documentation?]

VI. RESULTS

A. Baseline and closed-loop formulation

- Expectation and criterion: the LPV-LFR and original physics equations agree numerically across the payload range before model-error comparisons are meaningful
- Verdict and reason: report inertia invertibility, LFR/RK4 equivalence, and the remaining baseline error caused by the truth-only absorber
- Figure: baseline and oracle output errors versus time for representative payload positions; supports the claim that the omitted mode is systematic and position dependent
- Table: baseline free-run physical RMS for LPV and frozen-LTI models across held-out Y ; tests the research plan's Aspect 1 rather than assuming LPV superiority
- Expectation and criterion: direct and residual closed-loop forms match at numerical precision when controller assumptions hold
- Figure: recorded input, controller correction, and model error over one rollout; supports the interpretation that feedback can suppress visible output error while the open-loop model remains wrong

[**TODO:** The LPV-versus-frozen-LTI comparison is required by the plan but may not yet have a final matched result; run it or narrow Aspect 1 explicitly.]

B. Predictive benefit and additional-state contribution

- Expectation and criterion: augmentation must reduce held-out long-horizon error, not only the short training loss, relative to the same encoder-initialised baseline
- Verdict and reason: report the best validated improvement and the gap between window fit and free-run accuracy caused by integrated velocity bias
- Figure: validation RMS against optimiser updates for baseline and augmented arms, with window and 12 s metrics; supports or rejects objective alignment
- Figure: error spectra before and after augmentation with absorber pole and anti-resonance marked; tests whether the learned correction targets the missing band
- Figure: clamp each additional state and the full added-state vector to zero during rollout; quantifies what the trained latent dynamics add or harm
- Table: standard model, no-added-state ablation, zero-initialised latent path, and longer encoder history; connects structure changes to predictive benefit
- State the observed weak or harmful added-state contribution as a result if the final run does not reverse it, rather than inferring usefulness from state magnitude

[**TODO:** Are state Gramians useful for explaining each added state's contribution, or would intervention-based ablation be more direct and less linearisation-dependent?]

C. Parameter interpretability and orthogonality

- Expectation and criterion: OBC should reduce physical combination error without materially worsening held-out predictive fit relative to unconstrained joint estimation
- Verdict and reason: the completed arms reach 7.20% no projection, 7.69% tangent OBC, and 8.47% unfinished affine OBC from a common 9.49% start, so aggregate recovery does not establish OBC superiority
- Figure: validation fit versus combination error over training for all three arms; supports the claim that near-equal fit permits materially different physical parameters
- Figure: per-parameter error at matched budget; shows which combinations projection protects and which near-flat directions redistribute error
- Table: basis rank, condition, overlap, final and minimum parameter error, and validation RMS; links algorithm health to the recovery verdict
- Figure: ρ^* and predicted versus observed parameter bias by singular direction; supports the conditional result that true recovery is limited by non-orthogonal missing physics
- Explain why the 4.96% predicted floor accounts for part of the residual squared error but does not match the observed per-parameter pattern or prove convergence

[**TODO:** Should the incomplete affine arm be retained, rerun to a matched budget, or reported only as exploratory evidence?]

D. Generalisation and black-box comparison

- Expectation and criterion: improvement must progress from held-out position/phase interpolation to ASMP-

relevant motion, controller transfer, and quantified operating-range extrapolation without retuning

- Verdict and reason: separate completed simulation evidence from pending Telica results and explain any failure by excitation coverage, initialisation, or objective mismatch
- Figure: tracking overview plus a dedicated 200–400 ms settling zoom for baseline, augmented model, and black box on an approved operational trajectory
- Table: primary RMS, fixed-window settling RMS, parameter availability, and training data for baseline, augmented, and black-box models on every completed validation rung
- Figure or table: train with one controller and evaluate the frozen models with a credible changed controller; pair this with a plant-domain check for feedback masking
- Figure: error versus multisine amplitude, stroke, acceleration, scheduling position, or state-input distance from the training set; quantify extrapolation rather than calling all held-out data extrapolation
- Report the black box’s early improvement and later free-run degradation as a negative comparator result if no stable matched run supersedes it
- Aggregate the final learned-model comparison across neural initialization seeds and show a representative, not best, time trace; report phase realizations separately

[TODO: Freeze which validation rungs are complete enough for a final claim; omit rather than speculate about unfinished controller, noise, or hardware experiments.]

VII. DISCUSSION

A. Answers to the four research aspects

- Aspect 1: state whether explicit Y scheduling earns measurable predictive value over frozen LTI baselines
- Aspect 2: distinguish the expressive value of dynamic augmentation from the observed trainability of its added states
- Aspect 3: conclude that OBC enforces a unique local split but true parameter recovery is conditional on the real discrepancy’s orthogonality and optimiser convergence
- Aspect 4: compare simulation interpolation, motion-profile transfer, amplitude extrapolation, real-data evidence, and the black-box result without merging them into one score
- Treat controller transfer as the industrial digital-twin use case and as a closed-loop distribution shift distinct from operating-range extrapolation
- Interpretability matters when physical parameters support controller redesign or health monitoring, but its practical value must be tied to a named downstream decision
- Figure plan: introduce no new figure; interpret the architecture, OBC, and result figures already presented instead of adding a summary graphic

[TODO: Which result is the primary answer if predictive accuracy and parameter recovery favour different arms?]

B. Limitations and lessons

- Closed-loop feedback improves numerical trainability but can hide plant mismatch, suppress orthogonality overlap, and reward a model that is poor in open loop
- Short windows underweight velocity DC bias on free integrators; varying encoder and rollout length is therefore a structural sensitivity test, not routine tuning
- One strengthened absorber, currently noiseless simulation, a finite neural-seed campaign, and truth-only dry friction bound claims about nonlinear, noisy, and dissipative missing dynamics
- State the final seed count and corrected-friction evidence here; do not retain development-era caveats after the final campaign changes them
- Central-difference velocities, force calibration, and the train-validation-test coverage grid bound the credibility of any real-data conclusion
- The research plan expected validated real-machine generalisation; any missing Telica result is a departure to disclose, not replace with design documentation
- Augmentation versus residual-FRF fitting remains a trade between trajectory-domain nonlinear prediction and a simpler frequency-domain account of missing dynamics

[TODO: Should a cubic-spring benchmark, noise study, and Monte Carlo analysis be completed, or should the thesis explicitly narrow its claims to the measured linear absorber case?]

C. Implications and future work

- Design excitation so $J^\top \Delta^* = 0$ when true parameter recovery, rather than only a unique decomposition, is required
- Separate learning rates for physical and neural parameters and compare convergence against the predicted nonzero OBC bias floor
- Replace the short-horizon objective or couple windows so integrated velocity bias is visible during training
- Repair and rerun the Coulomb benchmark before claiming that the method learns a legitimate nonzero-mean residual
- Extend evaluation beyond the multisine range and use machine-relevant trajectories to delimit safe interpolation and extrapolation

[TODO: Which future-work item is a prerequisite for the thesis claim and which genuinely belongs after graduation?]

VIII. CONCLUSION

A. Conclusions

- Answer whether LPV-LFR dynamic augmentation improves dual-gantry settling prediction over the physics baseline and black-box comparator
- State whether the learned additional states contributed measurable missing dynamics under the final validated configuration
- Conclude that state-level OBC prevents local tangent overlap by construction, while true parameters are recovered

TABLE I: Nominal baseline parameters.

Parameter	Value	Parameter	Value
m_1	10.2 kg	m_2	10.7 kg
m_b	22.8 kg	m_h	10.1 kg
J_b	1.00 kg m ²	J_h	0.05 kg m ²
c_{g1}	14.5 N s/m	c_{g2}	20.3 N s/m
c_y	10.0 N s/m	d	0.10 m
c_{b1}	9.0 N m s/rad	c_{b2}	9.0 N m s/rad
k_{b1}	1987.5 N m/rad	k_{b2}	1987.5 N m/rad
L_b	0.725 m		

only when the physical discrepancy is orthogonal on the chosen excitation

- Give the final quantitative trade between prediction error and physical combination error on held-out data
- State whether the evidence covers relevant ASMPT motion, settling, controller transfer, or only a narrower simulation validation ladder
- Close with the practical recommendation for when ASMPT should use the augmented model and when the baseline or a simpler residual model is safer
- Figure plan: no conclusion figure; reserve the space for the final quantitative answer and limitations

[TODO: Rewrite every conclusion bullet after freezing the final result table; no planned or diagnostic result may appear as achieved.]

APPENDIX A

SUPPORTING DERIVATIONS AND REPRODUCIBILITY

A. Baseline and LFR details

- Give the logical-to-stage transformation, expanded inertia matrix, and proof of invertibility over the operating range
- List the compressed LPV-LFR matrices and internal signal dimensions needed to reproduce the baseline transition
- Provide the raw-to-identifiable parameter map and its Jacobian rank
- Figure (existing, optional): move the detailed coordinate and normalisation flow from `coordinates-normalisation-v1` here after checking derivative and scaling conventions

[TODO: Move these items into supplementary material if appendices count toward the 12-page limit.]

B. Identifiability of the physical parameters

Table I lists the nominal values used by the baseline. They are taken from the ETEL gantry reported by García-Herreros et al. [1, Table I]. Here θ collects the fourteen table entries other than the fixed length L_b .

The dynamics depend on θ only through the entries of M_0 , M_1 , M_2 , C and K in (2) and (5). Omitting zeros and symmetric

duplicates, these are

$$\begin{aligned}
 M_0 : m_1 + m_2 + m_b + m_h, \quad & \frac{L_b}{2}(m_1 - m_2), \\
 & J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2) + m_h d^2, \\
 & -m_h d, \quad m_h, \\
 M_1, M_2 : -m_h, \quad & m_h, \\
 C : c_{g1} + c_{g2}, \quad & \frac{L_b}{2}(c_{g1} - c_{g2}), \\
 & c_{b1} + c_{b2} + \frac{L_b^2}{4}(c_{g1} + c_{g2}), \quad c_y, \\
 K : k_{b1} + k_{b2}.
 \end{aligned}$$

Every entry is a function of φ in (13). Conversely, φ is recovered entry by entry:

$$\begin{aligned}
 m_h &= M_{0,33}, & d &= -M_{0,23}/m_h, \\
 m_\Sigma &= M_{0,11} - m_h, & m_\Delta &= 2M_{0,12}/L_b, \\
 J_{\text{eff}} &= M_{0,22} - m_h d^2, \\
 c_{g1}, c_{g2} : C_{11}, C_{12}, & c_y &= C_{33}, \\
 c_{b,\Sigma} &= C_{22} - \frac{L_b^2}{4}(c_{g1} + c_{g2}), & k_{b,\Sigma} &= K_{22}.
 \end{aligned}$$

The map from φ to the matrices is therefore injective. Twelve entries yield ten independent quantities because $M_{0,33}$, $M_{1,12}$ and $M_{2,22}$ all equal $\pm m_h$.

The matrices are fixed by input-output data. Since only the yaw coordinate has stiffness, every $q = [X_0 \ 0 \ Y_0]^\top$ with $\dot{q} = 0$ and $u_{\text{act}} = 0$ is an equilibrium. Linearizing (2) there, the term $(\partial M / \partial Y) \ddot{q}$ vanishes, so the small-signal model is $M(Y_0) \delta \ddot{q} + C \delta \dot{q} + K \delta q = P \delta u_{\text{act}}$ with transfer matrix $H(s) = P^\top Z(s)^{-1} P$, $Z(s) = M(Y_0) s^2 + C s + K$. As P is invertible, $Z(s) = P H(s)^{-1} P^\top$, whose polynomial coefficients give $M(Y_0)$, C and K . Three distinct Y_0 then give M_0 , M_1 and M_2 from the quadratic (5).

The four directions of θ that leave every entry unchanged are $\delta k_{b1} = -\delta k_{b2}$, $\delta c_{b1} = -\delta c_{b2}$, $\delta J_b = -\delta J_h$, and $(\delta m_1, \delta m_2, \delta m_b, \delta J_b) = t(-2/L_b^2, -2/L_b^2, 4/L_b^2, 1)$. Accordingly, the Jacobian of the twelve entries with respect to θ has rank ten for every positive θ , which was confirmed symbolically.

Positivity of the reduced coordinates alone does not guarantee that they correspond to a positive-definite inertia matrix. With $a = L_b m_\Delta / 2$, elimination of the m_h block gives

$$\min_Y \frac{\det M(Y)}{m_h} = (m_\Sigma + m_h) \left(J_{\text{eff}} - \frac{a^2}{m_\Sigma} \right).$$

Hence, for positive m_h , m_Σ , and J_{eff} ,

$$M(Y) \succ 0 \text{ for all } Y \in \mathbb{R} \iff m_\Sigma J_{\text{eff}} > \frac{L_b^2}{4} m_\Delta^2. \quad (14)$$

For the physical parameterization, the difference between the two sides is

$$\begin{aligned}
 m_\Sigma J_{\text{eff}} - \frac{L_b^2}{4} m_\Delta^2 &= m_\Sigma (J_b + J_h) \\
 &\quad + \frac{L_b^2}{4} [4m_1 m_2 + m_b(m_1 + m_2)],
 \end{aligned}$$

which is strictly positive for positive component masses and yaw inertias.

For completeness, let $R = M_0^{-1}$. The LTI matrices in (11) are

$$\begin{aligned} A &= \begin{bmatrix} 0 & I \\ -RK & -RC \end{bmatrix}, & B_w &= \begin{bmatrix} 0 & 0 \\ -RM_1 & -RM_2 \end{bmatrix}, \\ B_u &= \begin{bmatrix} 0 \\ RP \end{bmatrix}, & C_z &= \begin{bmatrix} -RK & -RC \\ 0 & 0 \end{bmatrix}, \\ D_{zw} &= \begin{bmatrix} -RM_1 & -RM_2 \\ I & 0 \end{bmatrix}, & D_{zu} &= \begin{bmatrix} RP \\ 0 \end{bmatrix}, \\ C_y &= [P^\top \quad 0]. \end{aligned}$$

Taking the Schur complement of the lower-right identity block gives

$$\det(I - D_{zw}\Delta(Y)) = \det(M_0^{-1}M(Y)) = \frac{\det M(Y)}{\det M_0}, \quad (15)$$

which proves the well-posedness equivalence used in Section II-B.

C. OBC and experiment details

- Derive the stacked projector, rank-deficient pseudoinverse case, epoch refresh, and additional-state zero extension
- Derive $J^+\Delta^*$ and list the measured singular-direction contributions to the predicted parameter bias
- List dataset identifiers, split membership, hyperparameters, seed, and exact metric definitions for every reported arm
- Figure (existing, optional): place `controller-chain-v1` here only if the controller reconstruction cannot be audited adequately in prose and equations

[TODO: Can code and data tables be cited through a repository or DOI instead of spending paper space on full reproducibility metadata?]

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